Ex. No : 03 Date :

Experiments on CISCO PACKET TRACER (Simulation Tools)

Aim :

To study the cisco Packe Tracer installation and user interface, and understand its environment for designing a simple network.

Introduction :

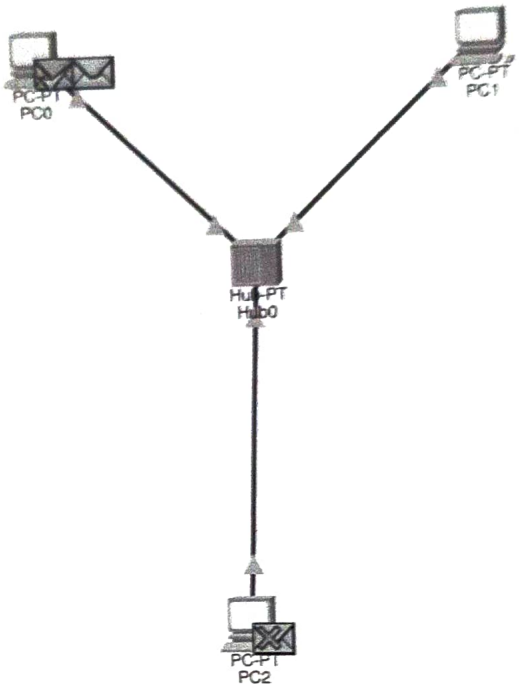
Packet Tracer is a Network simulation and modeling tool used to design and test networks without physical devices.

1. simulates network devices & environments.
2. Helps practice network configuration and troubleshooting.
3. Available for windows, Linux, Android and ios.
4. Protocols bahave similarly to real cisco hardware.

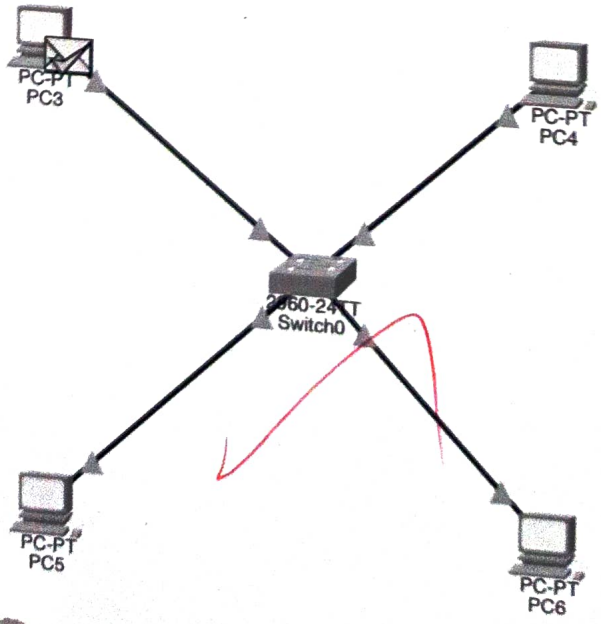
Installing Packet Tracer :

Download Packet Tracer from https://www.netacad.com using your cisco networking Academy account.

Hub


PC-PT
PC0
PC-PT
PC1
Hub-PT
Hub0
PC-PT
PC2

Switch


PC-PT
PC3
PC-PT
PC4
2960-24TT
Switch0
PC-PT
PC5
PC-PT
PC6

User Interface Overview :

1. Menu Bar - Open, Save, Print, Preferences, etc.
2. Main Toolbar - Shortcuts for open, save, zoom, undo, redo & networking information.
3. Logical / Physical Workspace Tabs - switch between logical & Physical views.
4. Workspace - Area for creating topologies and viewing simulations.
5. Common Tools Bar - Select, move, note, delete, inspect, resize & add PDU.
6. Realtime / Simulation Tabs - switch between realtime & simulation modes; control packet flow.
7. Network component Box
   * Device - Type Selection - select device categories.
   * Device - specific selection - Displays models of the selected category.
8. User - Created Packet Box - Create and view custom test packets.

Analyse the behaviour of network devices using CISCO PACKET TRACER Simulator:

1. Drag & drop:

* A Generic PCs & 1 Hub
* A Generic PCs & 1 switch.

2. select connections:

* choose copper straight-Trough cable.
* connect each PC to the Hub. Ensure the link LED turns green.
* similarly, connect the four PCs to the switch.

3. configure IP addresses:

* Open each PC → Desktop → IP configuration.
* Assign an IP address & subnet mask.
* Default Gateway and DNS are not required.

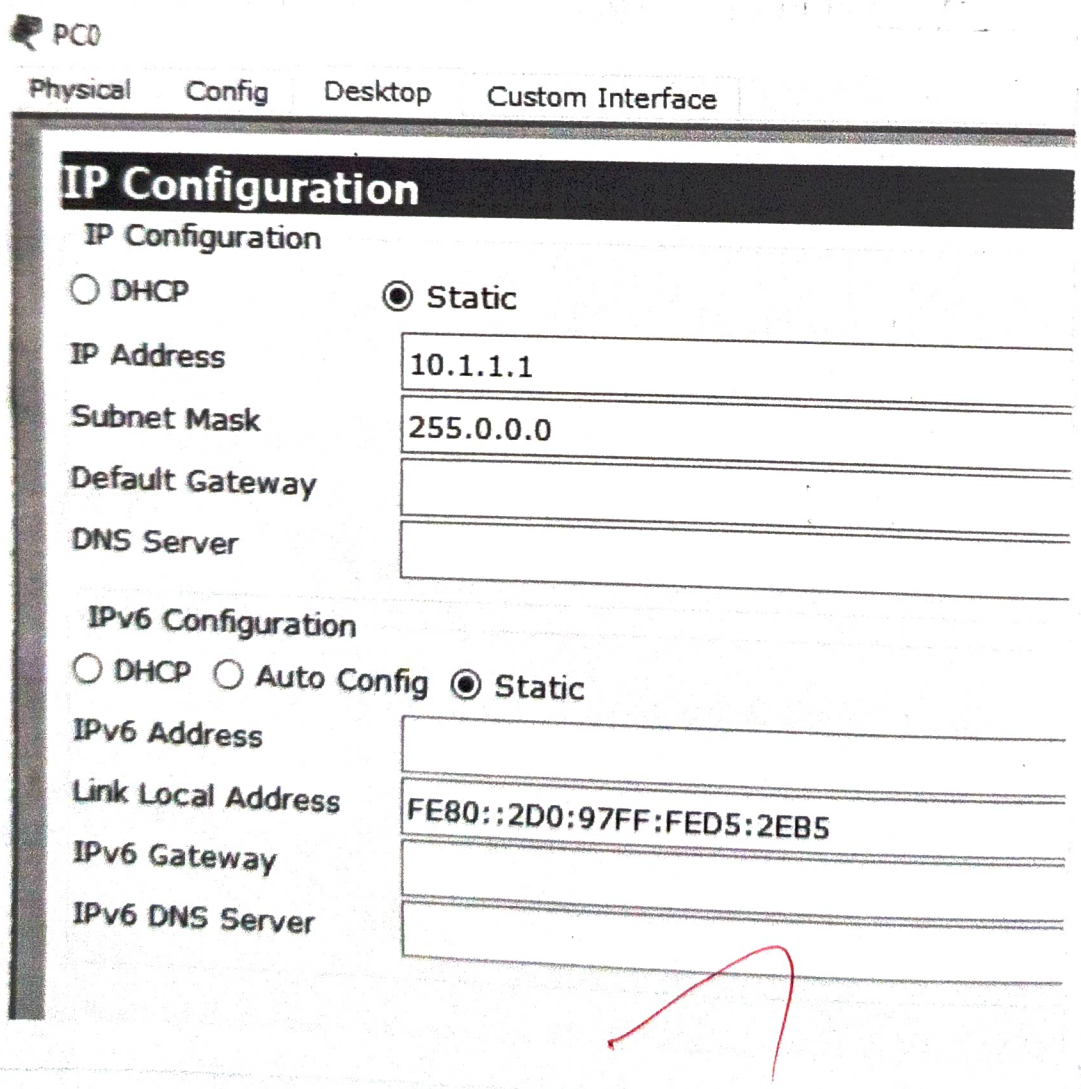
4. click Add simple PDU (message icon).

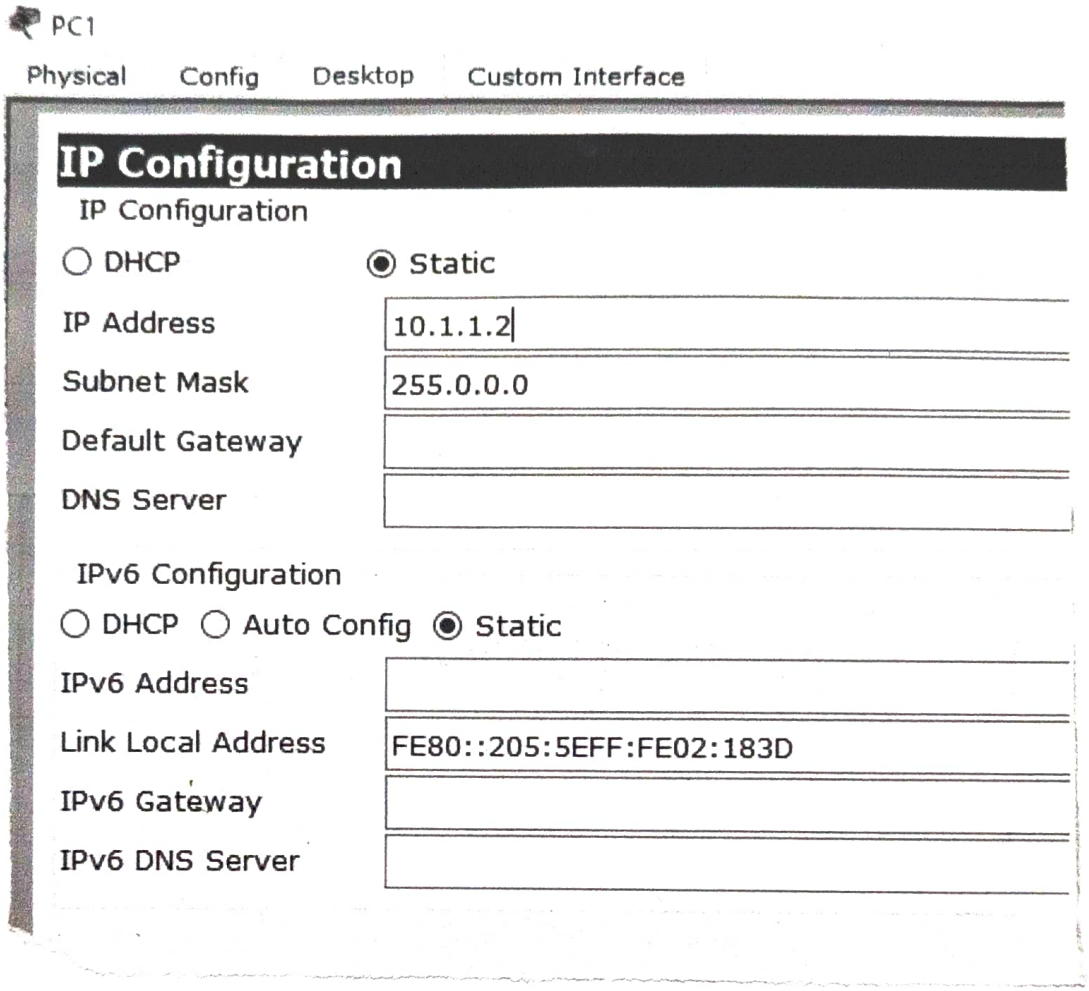
* Select the source PC and then the destination PC connected to the Hub.

5. In Realtime mode, observe the PDU & transmission.

6. Repeat the same steps for the PCs connected to the switch.

7. compare how the Hub & switch forwards the PDU, then write your observations and conclusion.





Result :

Thus the packet tracer tool installation and user Interface overview has studied and done successfully.